C29xMCAL_01.04.01 Release Notes

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Introduction

F29x MCAL is a cohesive set of tools for AUTOSAR based development on C2000 real-time controllers. It includes device-specific drivers (MCALs), EB Tresos Plugins for configuration, Bare-metal Examples for each driver.

F29H85x MCAL v01.04.01 is an early adopter release targeted for F29H85x.

Device Support

The following devices are supported with this release

-
  - MCU's
    - F29H859DU_Q1
    - F29H859TM_Q1
    - F29H859TU_Q1
    - F29P329SJ_Q1
    - F29P329SM_Q1
    - F29P589DM_Q1
    - F29P589DU_Q1
  - Hardware Platforms
    - F29H85X-SOM-EVM

What Is Supported

- MCAL / CDD (AUTOSAR **v4.3.1**):

Drivers	**Modules**
Micro controller Drivers	GPT, MCU, WDG
Communication Driver	CAN, LIN, SPI
I/O Drivers	DIO, PORT
Memory Drivers	FLS
Complex Device Drivers	ADC, XBAR, SENT, IPC, UART, PWM, ECAP, I2C

- Resource Allocator
- Reference Code:
  - Multi-Core startup code
- Build System:
  - CMake
  - CCS Projectspec

Note: The minimum version of CCS Theia required is CCSv20.03.00 or above and compiler version is C29Clang 2.0.0 STS. Please refer to the Dependencies section for the download links.

What Is Not Supported

- Multi-core MCALs as per AUTOSAR 4.3.1.

New In This Release

- MCAL / CDD :
  - Existing modules with bug fixes -
    1. CAN:
       1. Header inclusion structure is optimized as per AUTOSAR requirement
       2. Bug fix for MCAL-35513
       3. Bug fix for MCAL-35092
       4. Bug fix for MCAL-35093
    2. CDD ECAP:
       1. Header inclusion structure is optimized and bug fix for MCAL-35595
    3. BSW Stubs:
       1. Header inclusion structure is optimized and bug fix for MCAL-35594
    4. SPI:
       1. Header inclusion structure is optimized
    5. LIN:
       1. Header inclusion structure is optimized
    6. PORT:
       1. Bug fix for MCAL-35603
    7. WDG:
       1. Bug fix for MCAL-35603
    8. FLS:
       1. Header inclusion structure is optimized

Module Version Number

MCU	02.01.01
PORT	03.00.01
DIO	01.00.04
GPT	02.00.02
CAN	03.01.00
WDG	02.00.01
LIN	03.01.00
SPI	03.01.00
FLS	03.01.00
CDD ADC	03.00.01
CDD XBAR	03.00.01
CDD SENT	03.01.00
CDD PWM	03.01.00
CDD ECAP	03.01.00
CDD IPC	02.01.00
CDD UART	03.00.01
CDD I2C	01.01.00

Note:

We follow Autosar's requirement [SRS_BSW_00321] for versioning of each module-

5.2.4.10 [SRS_BSW_00321] The version numbers of AUTOSAR Basic Software Modules shall be enumerated according specific rules

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Type:	Valid
Description:	The version numbers of AUTOSAR Basic Software Modules shall be enumerated according to the following rules: - Increasing a more significant digit of a version number resets all less significant digits - The PATCH_VERSION is incremented if the module is still upwards and downwards compatible (e.g. bug fixed) - The MINOR_VERSION is incremented if the module is still downwards compatible (e.g. validfunctionality added) - The MAJOR_VERSION is incremented if the module is not compatible any more (e.g. existing API valid)
Rationale:	Provide unambiguous version identification for each module, provide version cross check as well as basic version retrieval facilities. Compatibility is always visible!
Use Case:	Example: ADC module with version 1.14.2:

Fixed In This Release

Key	Summary	Component/s	Severity	P
MCAL-35603	PORT & WDG: PortNumberOfPortPins and WdgMaxTimeout doesn't have correct fallback value	Port, Wdg	S3-Minor	
MCAL-35595	Missing Inclusion of os.h in ECAP Header Structure	CDD ECAP	S2-Major	
MCAL-35594	Rename SSU Macro in os.h to Resolve Conflict with C29 SDK	Common	S3-Minor	
MCAL-35593	Move hw_memmap.h Inclusion from Os.h to Generated Config File	Can	S2-Major	
MCAL-35513	When CanRxProcessing is INTERRUPT and CanTxProcessing is POLLING, interrupts are not processed	Can	S2-Major	
MCAL-35092	CAN: CanIf_TriggerTransmit should be called only if enabled	Can	S3-Minor	

6 issues

Known Issues And Limitations

There are few limitations and known issues associated with this release

Limitations

- CDD ECAP -
  - In case of high frequency signals (i.e. 500Khz and above), software overhead in user callback shall be kept to minimum to avoid missing edges.
- Test Infrastructure : Following tests have not been performed due to limitations of available test infrastructure -
  - CDD SENT : Following modes in SENT are not tested -
    - Enhanced 16-bit Slow Mode.
    - Nibble Count 1,2,3,4,5.
    - MTP Mode with all 4 sensors connected.
  - CDD IPC : IPC is validated between CPU1 and CPU3. Since, CPU2 is configured in lockstep in our framework; IPC between CPU1 and CPU2 or CPU2 and CPU3 is not validated.

  - Fls : Timeout tests are not performed as FLASH characterization data is not yet available for this device.
  - Spi : Board to board Spi communication validated upto 40MHz. 50MHz have been only test in loopback mode.
  - CDD I2C: Target mode is not supported.
- Compiler
  - Link Time optimization is currently not supported.

New Known Issues

Key	Summary	Component/s	Workaround	Severity

No issues found

Existing Known Issues

Key	Summary	Component/s	Workaround	Severity
MCAL-35347	I2C: Cancel API is not functional	CDD I2C	Do not cancel the ongoing transmission using API. Instead handle this situation in application code.	S2-Major
MCAL-32321	All Modules: Link time optimization is not verified for all modules	All	By default LTO is disabled, user shall not enable it.	S2-Major

2 issues

Quality Status

Sr. No.	Work Product	Availability	Where to find Artifacts	Current Quality Status
1	Dynamic Analysis Report 1. MCDC 2. Branch 3. Region 4. Line 5. Function	No	CSP	Will be available in future releases. Please refer to device roadmap for details.
2	Static Analysis Report 1. Static 2. HIS 3. MISRA-C	Yes	Installer:*\docs\Static_Analysis	All Mandatory & Critical warnings and errors are resolved. Required and Advisory warnings will be fixed in future releases.
3	Test Reports: Test Cases and Test Results	Yes	Installer:*\docs\Test_Report	All test reports are present in installer.
4	Bi-directional Traceability Report 1. MRD↔SPS 2. SPS↔SW Arch ↔SW Design Doc↔SW Units (Code) 3. MRD↔SPS↔Test Case 4. Test Case↔Arch Test Case↔Design	No	CSP	Will be available in future releases. Please refer to device roadmap for details.

5	Customer Docs: User Guide	Yes	Installer:*\docs\User_Guide	User Guide is present in installer as well as softwaredl
6	SW FMEA	No	CSP	Will be available in future releases. Please refer to device roadmap for details.
7	Detailed Design Document	No	CSP	Will be available in future releases. Please refer to device roadmap for details.
8	SW Manifest	Yes	Installer:*\mcal_manifest	SW manifest is present in installer
9	Software Product Specification: List of functional, safety requirements for SW components	No	CSP	Will be available in future releases. Please refer to device roadmap for details.
10	Architecture Document	No	CSP	Will be available in future releases. Please refer to device roadmap for details.

11	SW Safety Manual	No	CSP	Will be available in future releases. Please refer to device roadmap for details.
12	Safety Assessment	No	CSP	Will be available in future releases. Please refer to device roadmap for details.

Dependencies

This release has dependency on the following tools

- CCS v20.03.00 or above version.
- C29Clang Compiler 2.0.0 STS.
- EB Tresos (ACG 8.8.12) : Request access here.

References

Technical Support

https://e2e.ti.com/support/microcontrollers/c2000/f/171